



ENGINEERING PROJECT MANAGEMENT

Lecture # 1

What is Project?

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Faculty Profile

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Qualifications:

- ▶ Postdoc Industrial Sustainability - PSU Thailand
- ▶ PhD in Sustainability - PSU Thailand, CYCU Taiwan
- ▶ MS Engineering Management - CUST Islamabad
- ▶ BS Electronic Engineering – MAJU Islamabad

Experience:

5 years experience as a Lecturer at CUST Islamabad

Course Information

- Course Name - Engineering Project Management
- Course Code - OTM 455
- Enrollment Code - 153789406
- Credit Hours - 2
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- Visiting Hours - 2-4 PM (Tuesday & Thursday)
- Location - 1st Floor, Department of Engineering Management

Why Engineers should learn about PM?

- Project Management - Preparing for success
- Being applied for 60 years
- 40% of the projects are unsuccessful - even more for the larger projects
- Hard to manage
 - Front end development
 - Building and maintaining relationships in teams
 - Extended teams (owner/client, contractors, suppliers and users)
- People are the key

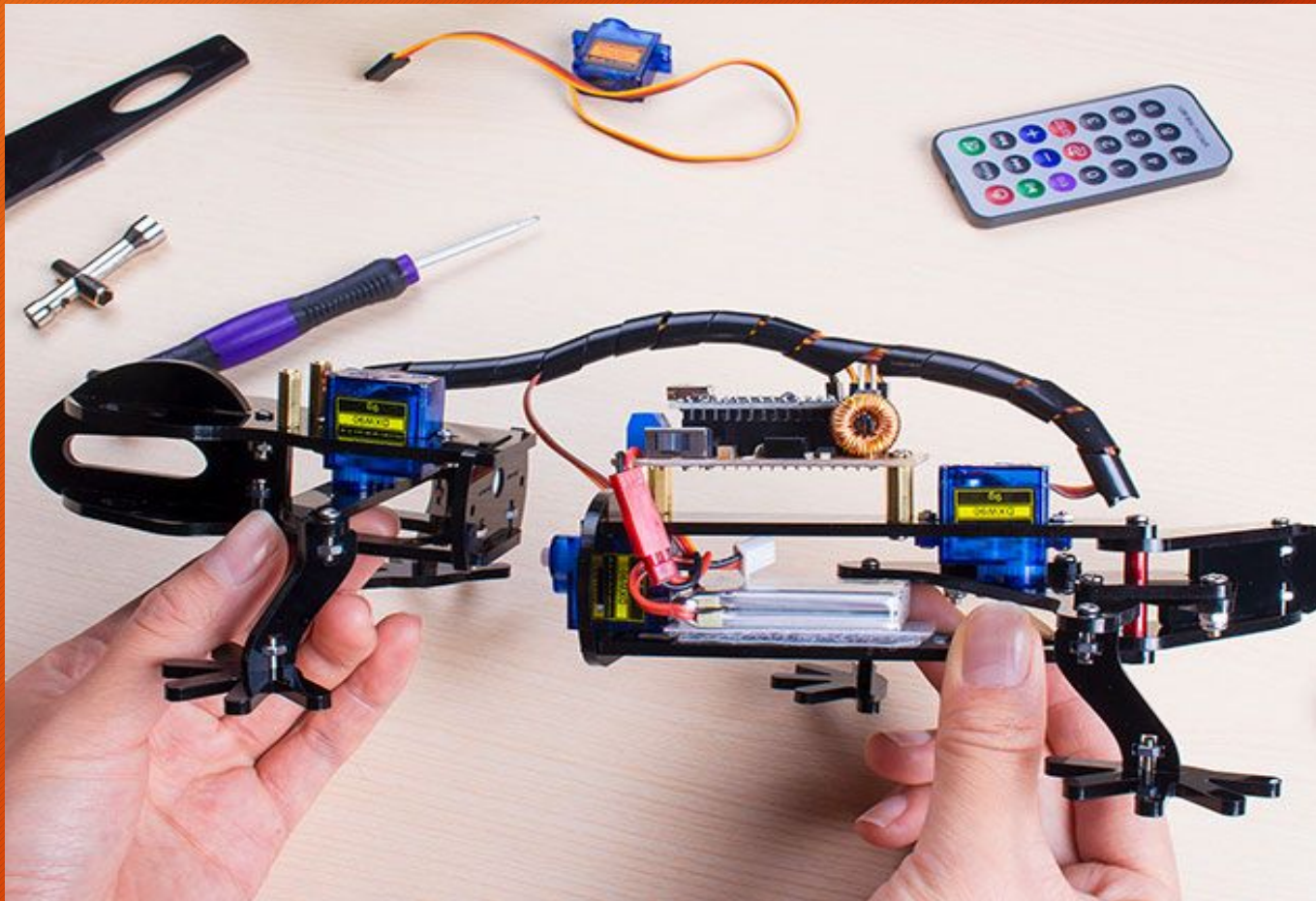
Project - A Bridge



Project - A Plant



Project - Tech Project



Similarities b/w all projects

- Certain constraints
- Multidisciplinary teams
- High levels of uncertainty
- Set your budget and involve a team of specialists to make your dream come true (....and not end-up in a nightmare....)

What is a Project?

- *A project is an endeavor in which human, financial and material **resources** are organized in a novel way to undertake a unique scope of work, of given specifications, within constraints of cost and time, so as to achieve the beneficial change defined by qualitative and quantitative objectives.*

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Prof. Turner

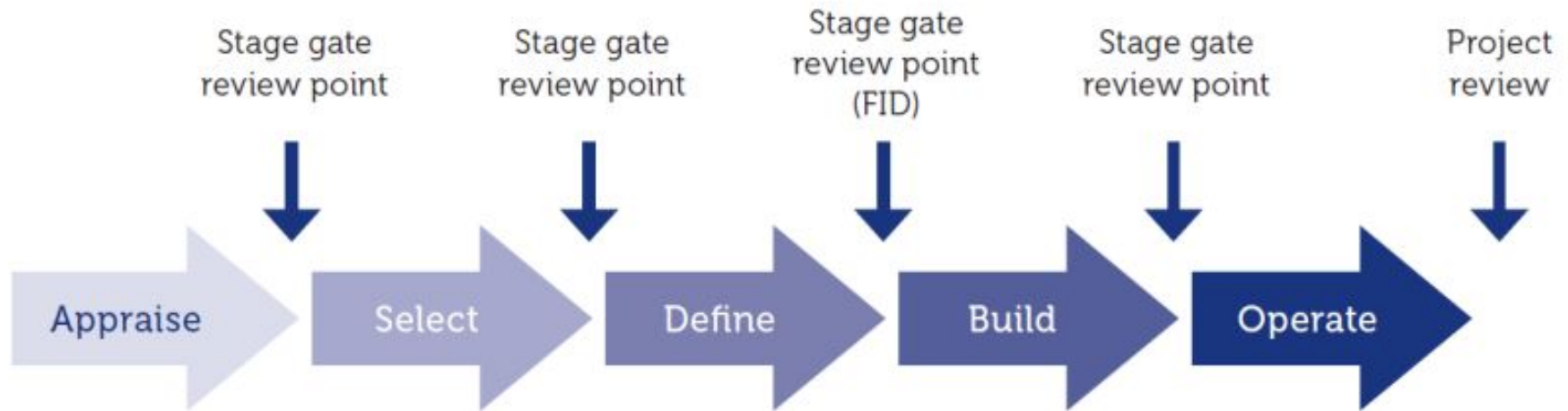
Project Management

- In essence, a project is a temporary organization, brought to life to deliver a unique scope, within the given boundary conditions.

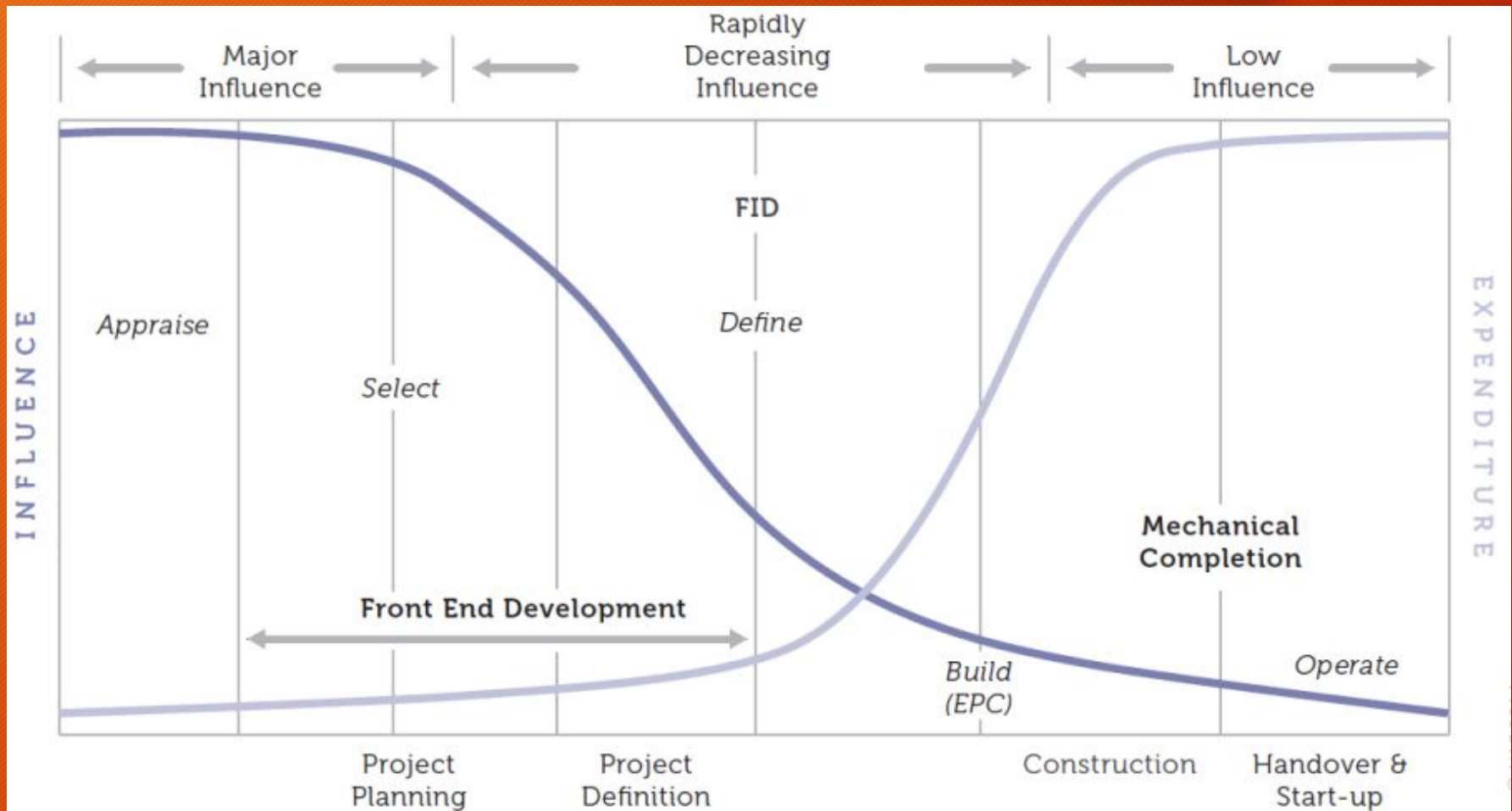
Project Life Cycle Phases



Project Phases (In General)

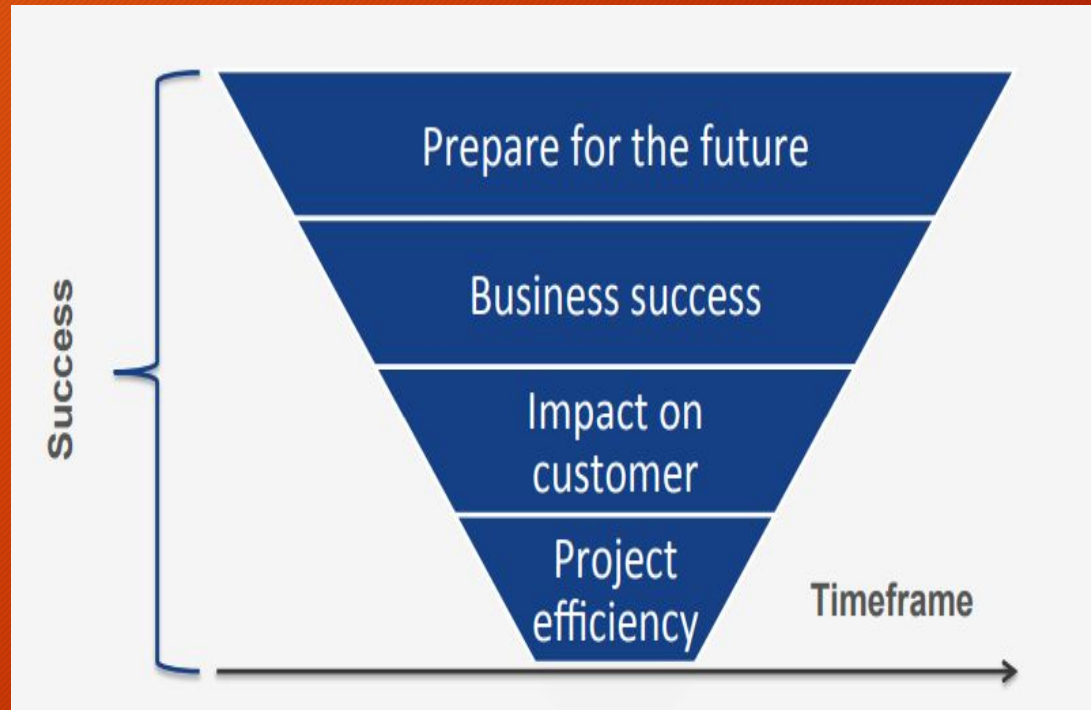


Final Investment Decision - FID

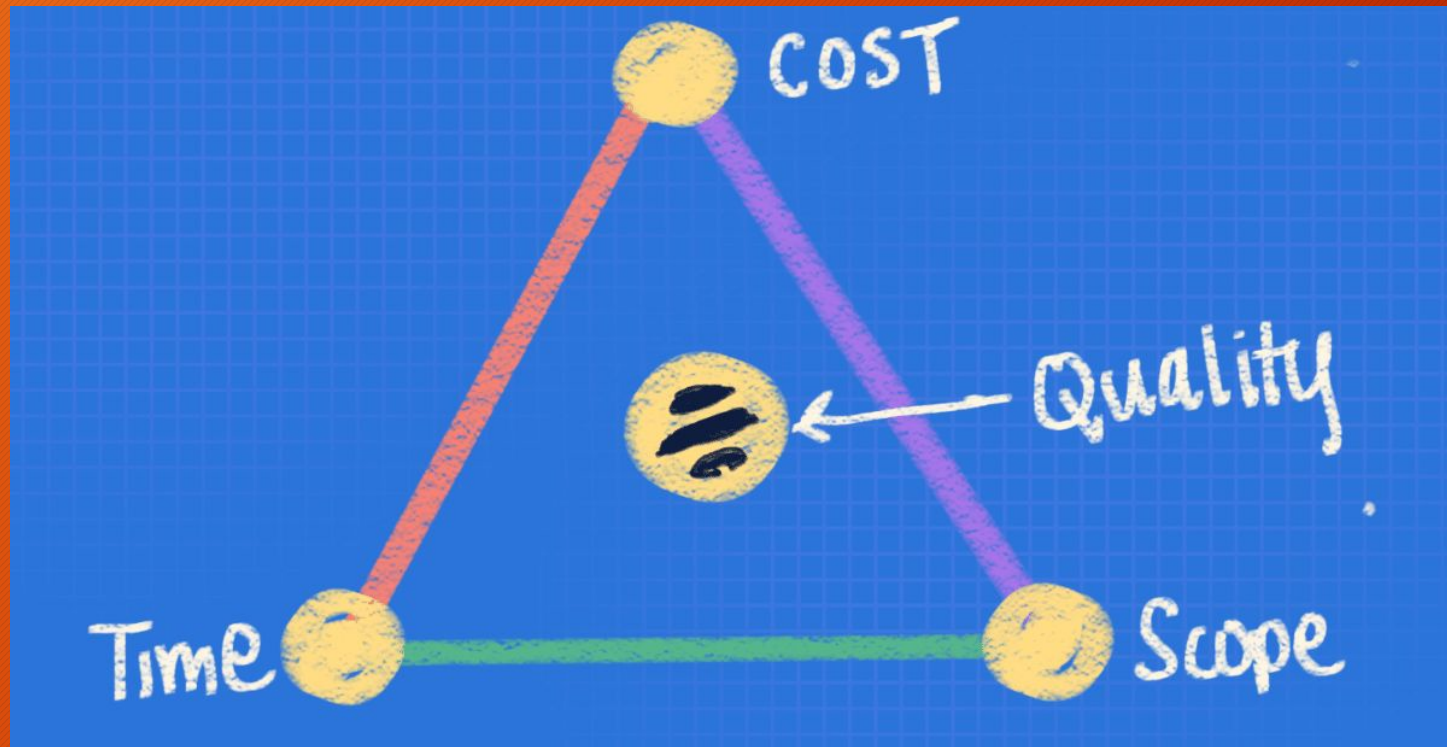


Project Success

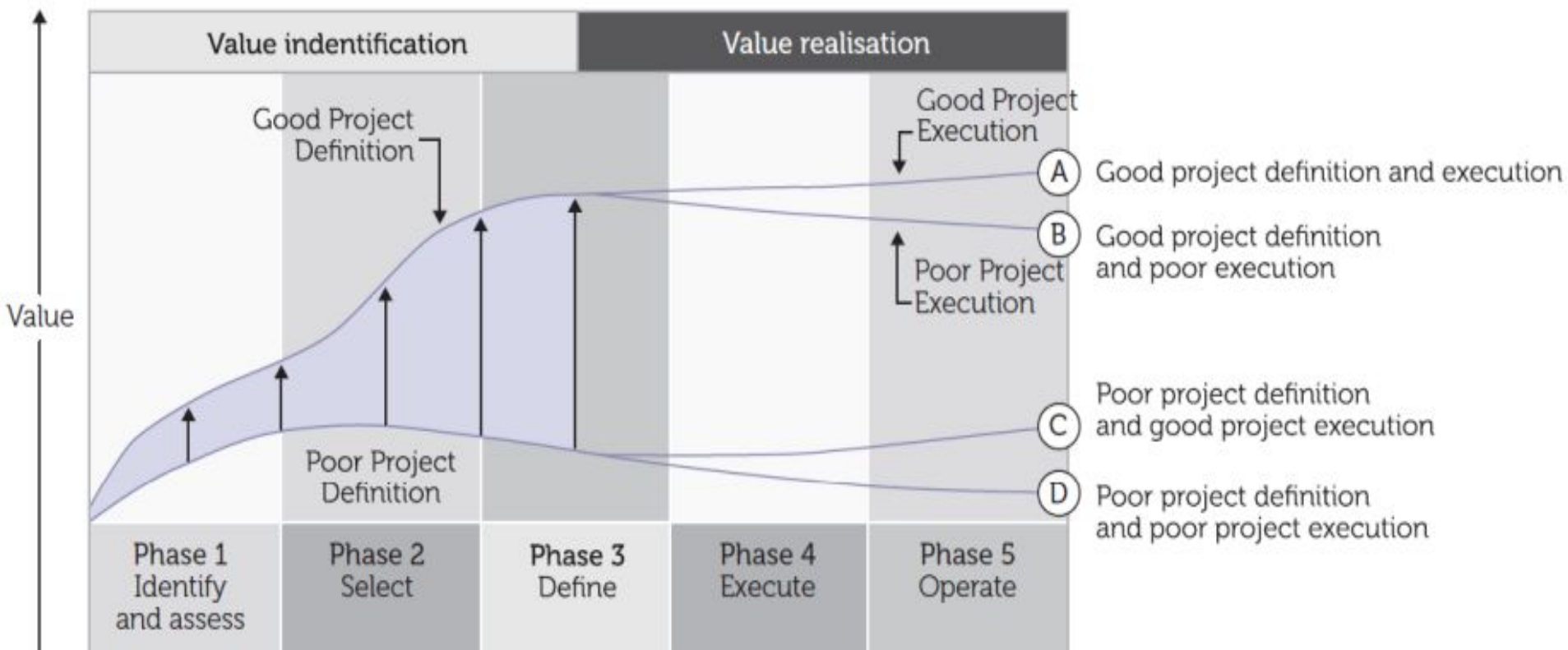
- Prepare for the future
- Business success
- Impact on customer
- Project efficiency



Project Success - Iron Triangle



Value Management



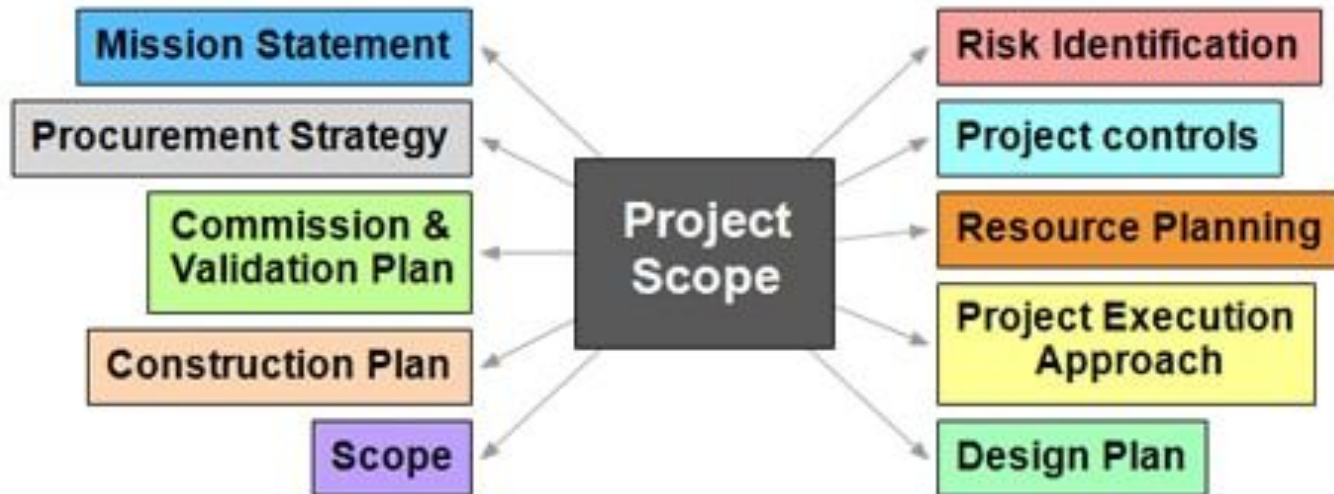
Selecting the Right Project

- Strategic Fit
- Managing the portfolio
- Business case central



Project Scope

What to include in your project scope statement:



Planning ?



Planning



“Planning is bringing the future into the present so that you can do something about it now.”

“He who fails to plan, plans to fail.”

“Plans are nothing; planning is everything.”

Dwight D. Eisenhower

- The process of setting goals, developing strategies, and outlining tasks and schedules to accomplish the goals.

Discussion

